LOTQ-0031 / BACKTESTING / INTERMEDIATE / FIELD GUIDE

Backtesting Biases and Research Errors

Audit causal, economic, statistical, and operational defects in historical research.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0031
CATEGORY	Backtesting
DIFFICULTY	Intermediate
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

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How to use this manual

INTERMEDIATE

1. BEGIN WITH THE EARLIEST BOUNDARY

Trace an attractive return backward through NAV, holdings, fills, orders, decisions, and source records. Stop at the first impossible or unreconciled transition because later statistics cannot repair it.

2. SEPARATE INVALIDITY FROM UNCERTAINTY

Lookahead, survivor-only data, and fantasy fills invalidate the tested question. Sampling variation and regime uncertainty may leave the question valid but the answer imprecise.

3. PRICE THE FULL RESEARCH SEARCH

Include abandoned models, parameters, universes, filters, charts, and holdout peeks in the selection history. Judge the procedure that produced the winner, not only its final specification.

4. RECONCILE ECONOMICS BEFORE METRICS

Require every quantity change, cash movement, fee, action, and liability to balance in the ledger. Compute returns and attribution only after state and NAV identities pass.

5. STRESS INTERACTING ASSUMPTIONS

Vary delay, spread, impact, participation, borrow, stale inputs, regimes, and capital jointly where they reinforce one another. Preserve complete surfaces instead of a few favorable scenarios.

6. PRESERVE NEGATIVE KNOWLEDGE

Store failed variants, invalidated claims, defect causes, and permanent regression tests. A repaired bug should become an institutional control rather than a forgotten anecdote.

DECISION STANDARD

Promote only when evidence is point-in-time, simulation is executable and reconciled, selection is disclosed, validation is independent, implementation margin survives stress, and live parity is governed.

REFERENCE SHEET

Notation and audit contract

INTERMEDIATE

I_T AND U_T

Information legally available and the eligible point-in-time universe at decision t. Vintages, identity, age, membership, quality, and borrow state are explicit.

D_T, O_T, F_T, Q_T

Decision, submitted order, realized fill, and held quantity. They are separate states connected by causal event IDs.

NAV_T AND R_T

Net asset value and return after external-flow adjustment. Cash, positions, receivables, liabilities, fees, and actions reconcile exactly.

THETA AND M

The candidate specification and the research family from which it was selected. M includes viewed manual choices, not only formal grid points.

ALPHA AND FWER

Per-test significance level and family-wise error rate. Under independent nulls, FWER = 1 - (1-alpha)^M.

RHO_K AND N_EFF

Lag-k dependence and effective sample size. A rough approximation divides row count by 1 + 2 times the sum of relevant autocorrelations.

C(Q, STATE)

Implementation burden conditional on order size and market state. Spread, fees, impact, delay, financing, borrow, and opportunity loss belong in the function.

V_K AND PHI_K

Dataset vintage and fold-scoped fitted transform. Both are immutable artifacts bound to a run and decision boundary.

MINIMUM AUDIT RECORD

Bind hypothesis, search lineage, code, parameters, data vintages, calendars, environment, random seeds, simulation events, metrics, reviews, and approvals.

PROCESS FLOW

Bias-resistant research workflow

INTERMEDIATE

1. **Register the question**
   Mechanism, decision use, falsifiers, primary metrics, search family, and stopping rule.

2. **Freeze lawful evidence**
   Identity, vintages, availability, membership, actions, calendars, quality, liquidity, and borrow.

3. **Build isolated folds**
   Causal joins, fold-fitted transforms, labels, purge, embargo, and sealed evaluation roles.

4. **Simulate economic state**
   Decision, target, order, queue, fill, holding, cash, action, liability, and valuation.

5. **Reconcile and benchmark**
   Ledger identities, costs, financing, exposures, matched baselines, and attribution residual.

6. **Price the search**
   Trial registry, dependence, uncertainty, complete specification surfaces, and replication.

7. **Stress implementation**
   Delay, spread, impact, capacity, borrow, outages, common exit, and joint failure states.

8. **Govern live parity**
   Immutable artifacts, shadow replay, monitoring, incident tests, rollback, and retirement.

HARD-STOP RULE

Future knowledge, survivor-only rosters, unreconciled NAV, impossible fills, hidden search, exhausted holdouts, unsupported costs, or absent rollback blocks promotion regardless of headline performance.

Research error taxonomy

INTERMEDIATE

ONE-SENTENCE MEANING

A research error is any defect that makes historical evidence answer a different question from the one capital will face. Errors can enter data, timing, simulation, statistics, interpretation, or governance.

MEMORY JOGGER

Classify the defect before repairing the result. An upstream causal error invalidates downstream precision, even when the final chart looks plausible.

WHY IT MATTERS IN QUANT RESEARCH

A taxonomy turns vague skepticism into testable controls and assigns each failure to an owner. It also prevents a statistical adjustment from being used to excuse impossible data or execution.

FORMULA INTUITION

Observed performance = economic effect + sampling noise + selection optimism + implementation error. The terms interact, so the decomposition is diagnostic rather than directly observable.

SIMPLE MARKET EXAMPLE

A high-Sharpe small-cap strategy may combine survivor-only data, optimistic fills, and a lucky parameter. Correcting only transaction costs still leaves the research claim invalid.

PYTHON / SYSTEM IMPLEMENTATION

Tag every test and incident by source, boundary, severity, detectability, and affected decisions. Store the earliest invalid event so reruns can prove the defect no longer propagates.

CONFUSIONS TO AVOID

Bias is not synonymous with fraud or carelessness. Many errors arise from convenient defaults that silently change the historical information set or feasible action set.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The taxonomy links data quality, feature lineage, simulation, validation, experiment tracking, promotion, and incident review. Control coverage should be reported by error class.

RELATED CONCEPTS AND QUICK RECALL

Related: causal validity, sampling error, model risk, operational risk; together they separate wrong evidence from uncertain evidence. Recall task: Take one attractive backtest and name a data, timing, execution, statistical, and governance error that could explain it.

The frozen strategy contract

INTERMEDIATE

ONE-SENTENCE MEANING

A strategy contract fixes the hypothesis, information set, universe, rule, portfolio mapping, execution, costs, and evaluation before results are judged. It makes discretionary repairs visible as new experiments.

MEMORY JOGGER

If a second researcher cannot replay the procedure without asking questions, the test is underspecified. Every unresolved default is a hidden degree of freedom.

WHY IT MATTERS IN QUANT RESEARCH

The contract separates economic reasoning from choices made after seeing outcomes. It also creates a stable unit for comparison, approval, monitoring, and retirement.

FORMULA INTUITION

Procedure theta = {U, I, feature, target, constraints, schedule, execution, costs, accounting, metrics}. A change to any element creates a new candidate with a new search history.

SIMPLE MARKET EXAMPLE

Changing a missing-value rule after a drawdown can remove the bad trades without changing the strategy name. The contract records that repair as a separate version requiring fresh evidence.

PYTHON / SYSTEM IMPLEMENTATION

Store a machine-readable manifest with typed parameters, clock definitions, dataset vintages, code hashes, seeds, and metric contracts. Fail closed when a field is implicit or mutable.

CONFUSIONS TO AVOID

A notebook narrative is not a contract because cell order, cached state, and informal choices remain ambiguous. A frozen file alone is insufficient if its queries retrieve changing history.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The contract is the interface between hypothesis registry, data catalog, backtest engine, experiment tracker, and model registry. Live orders should resolve to one approved contract ID.

RELATED CONCEPTS AND QUICK RECALL

Related: preregistration, manifests, semantic versioning, change control; these controls preserve the tested decision rule. Recall task: List the contract fields that must change when a daily signal moves from next-open to close-auction execution.

FOUNDATIONS

Causal evidence chain

INTERMEDIATE

ONE-SENTENCE MEANING

A causal evidence chain traces each reported return backward through NAV, holdings, fills, orders, decisions, and point-in-time inputs. The first impossible link determines where the backtest stops being evidence.

MEMORY JOGGER

Audit from capital back to bytes. A valid statistic cannot rescue a return produced by a position the strategy could not have held.

WHY IT MATTERS IN QUANT RESEARCH

Chain-based review localizes defects and prevents teams from debating aggregate Sharpe before reconciling mechanics. It also provides the structure for replay and live parity tests.

FORMULA INTUITION

Return_t <- NAV_t <- holdings_t <- fills_t <- orders_t <- decision_t <- available data_t. Each arrow has timing, identity, and conservation invariants.

SIMPLE MARKET EXAMPLE

A profitable day is invalid when the ledger shows 500 shares but fills sum to 300 and no corporate action occurred. The error exists before any risk-adjusted metric is computed.

PYTHON / SYSTEM IMPLEMENTATION

Assign immutable IDs to source records, decisions, orders, fills, positions, and ledger entries. Build lineage queries that reconstruct one return and fail when an edge is missing.

CONFUSIONS TO AVOID

Lineage is not merely a list of filenames. It must preserve transformation state, availability clocks, and the exact economic events that changed exposure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evidence graph connects ingestion, feature computation, portfolio construction, execution, accounting, metrics, and governance. Incident response should begin at the earliest divergent node.

RELATED CONCEPTS AND QUICK RECALL

Related: event sourcing, double-entry accounting, data lineage, causal replay; all support return-level auditability. Recall task: Choose one portfolio return and describe the minimum records required to reproduce it from raw evidence.

Hidden degrees of freedom

INTERMEDIATE

ONE-SENTENCE MEANING

Hidden degrees of freedom are choices the researcher can vary without recording them as formal trials. They include windows, filters, charts, endpoints, data repairs, and when to stop searching.

MEMORY JOGGER

Count choices, not just model parameters. A verbal story can overfit when many definitions were tried and only one was retained.

WHY IT MATTERS IN QUANT RESEARCH

Unlogged flexibility makes selection optimism impossible to estimate and encourages accidental narrative fitting. Recording choices reveals the true research family behind the winner.

FORMULA INTUITION

Effective search size depends on the number and dependence of attempted paths, not only the final grid. Correlated choices still increase the chance of discovering a favorable accident.

SIMPLE MARKET EXAMPLE

A researcher tries three universes, four horizons, two cost rules, and several crisis exclusions but logs one final run. The apparent single hypothesis actually emerged from dozens of paths.

PYTHON / SYSTEM IMPLEMENTATION

Capture every executed configuration automatically and register material manual decisions with timestamps. Link child experiments to a parent hypothesis so abandoned paths remain visible.

CONFUSIONS TO AVOID

Informal inspection is still a trial when it influences the final specification. Deleting failed code does not delete its effect on selection.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Search lineage belongs beside results in the experiment tracker and promotion packet. Reviewers need both the winning artifact and the route by which it was chosen.

RELATED CONCEPTS AND QUICK RECALL

Related: garden of forking paths, researcher discretion, experiment registry, selection bias; these describe unpriced flexibility. Recall task: Estimate the hidden search behind one published strategy description and identify which choices materially affect P&L.

Availability and vintage leakage

INTERMEDIATE

ONE-SENTENCE MEANING

Availability leakage occurs when a decision uses a value or revision that was not accessible at its cutoff. Vintage leakage replaces what was known then with a later corrected history.

MEMORY JOGGER

The date described by a record is not the date the strategy knew it. Preserve each knowledge interval, not only the final value.

WHY IT MATTERS IN QUANT RESEARCH

Final-vintage data often looks cleaner and more predictive because errors and uncertainty were repaired later. Using it exaggerates both signal quality and operational reliability.

FORMULA INTUITION

A record is admissible only when availability_time <= decision_time and its vintage was current at that moment. Later restatements start new intervals rather than rewriting earlier decisions.

SIMPLE MARKET EXAMPLE

A quarterly value filed in May and restated in August cannot inform an April rebalance. Joining by fiscal period end leaks both the first release and the later correction.

PYTHON / SYSTEM IMPLEMENTATION

Store event time, publication time, vendor arrival, ingestion time, effective interval, and source version. Add future-mutation tests proving later records cannot alter earlier outputs.

CONFUSIONS TO AVOID

Applying shift(1) does not repair final-vintage fundamentals or retroactive classifications. Row order is not a substitute for real availability semantics.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Vintage-aware storage feeds as-of joins, feature building, replay, fold construction, and incident analysis. The run manifest binds the exact source snapshot.

RELATED CONCEPTS AND QUICK RECALL

Related: point-in-time data, bitemporal tables, as-of joins, restatements; together they define lawful historical knowledge. Recall task: Draw the knowledge intervals for an original filing and a later restatement, then identify valid decisions for each.

DATA AND CLOCKS

Survivorship and universe selection

INTERMEDIATE

ONE-SENTENCE MEANING

Survivorship bias tests historical decisions on assets that survived until a later selection date. Universe selection bias also enters when eligibility depends on future completeness, liquidity, or outcomes.

MEMORY JOGGER

Reconstruct who could have been selected before calculating ranks or features. Keep failures, entrants, deletions, and temporary ineligibility in the opportunity set.

WHY IT MATTERS IN QUANT RESEARCH

Removing failed names inflates quality, liquidity, return, and apparent signal breadth. It also distorts cross-sectional transforms because every rank depends on the roster.

FORMULA INTUITION

U_t contains assets identifiable and eligible using information available at t. Every feature rank, benchmark, portfolio constraint, and support count is conditional on that historical set.

SIMPLE MARKET EXAMPLE

Using today's index members for a fifteen-year test excludes bankrupt firms and includes later winners before admission. A historical constituent file restores the actual choice set.

PYTHON / SYSTEM IMPLEMENTATION

Use effective-dated security masters, listing histories, membership events, identifier maps, and reason-coded eligibility. Persist roster snapshots or hashes at each decision.

CONFUSIONS TO AVOID

Filtering securities with short future histories uses outcome information even when all price features are lagged. A vendor symbol list can still be survivor-only.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Universe construction precedes feature computation and feeds benchmark, borrow, capacity, and risk services. Monitoring should separate signal turnover from roster turnover.

RELATED CONCEPTS AND QUICK RECALL

Related: selection bias, delisting bias, historical constituents, security master; these determine the feasible cross-section. Recall task: Explain how deleting incomplete histories changes both asset returns and the percentile rank of every survivor.

Terminal events and corporate actions

INTERMEDIATE

ONE-SENTENCE MEANING

Terminal events and corporate actions must change prices, shares, cash, identity, and tradability consistently. Omitting difficult exits or double-adjusting actions creates artificial wealth.

MEMORY JOGGER

A missing quote is not a harmless zero, and a delisting is not an optional row. Preserve the economic event through the portfolio ledger.

WHY IT MATTERS IN QUANT RESEARCH

Small numbers of bankruptcies, mergers, splits, and special distributions can dominate a long backtest. Correct handling is essential for tails, shorts, and small-cap strategies.

FORMULA INTUITION

NAV after an action equals lawful cash plus adjusted quantity times lawful mark plus receivables minus liabilities. Price and share transformations must conserve value before event-specific proceeds.

SIMPLE MARKET EXAMPLE

A 2-for-1 split halves raw price and doubles quantity without creating loss. A later cash delisting requires proceeds or a conservative recovery, not forward-filled value forever.

PYTHON / SYSTEM IMPLEMENTATION

Ingest action events with ex, record, pay, and effective dates; reconcile quantity and cash changes. Test transformed series against independently calculated total returns.

CONFUSIONS TO AVOID

Backward-adjusted prices can embed later knowledge and should not also receive duplicate cash dividends. Splicing successor identifiers may fabricate an asset that never existed.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The action service supplies feature views while the ledger consumes actual quantity and cash events. Both must reference the same event version and identity map.

RELATED CONCEPTS AND QUICK RECALL

Related: delisting returns, split factors, dividends, identifier continuity; these preserve economic history across structural changes. Recall task: Reconcile shares, price, cash, and NAV through a split, dividend, merger, and zero-recovery delisting.

Calendar alignment and stale observations

INTERMEDIATE

ONE-SENTENCE MEANING

Calendar bias appears when records are aligned by labels rather than economic sequence and availability. Stale observations can create false lead-lag, low volatility, and impossible cross-market trades.

MEMORY JOGGER

Same date does not mean same information time. Track venue sessions, time zones, age, and next executable window.

WHY IT MATTERS IN QUANT RESEARCH

Cross-asset research is especially vulnerable because closes, holidays, and auctions occur asynchronously. Incorrect alignment can manufacture prediction that disappears under lawful clocks.

FORMULA INTUITION

An as-of join selects the latest record available by the cutoff and records age = cutoff - source_time. Eligibility and feature logic should reject observations beyond a declared maximum age.

SIMPLE MARKET EXAMPLE

A US close cannot react to an Asian close that occurs later in absolute time, even if both rows share a calendar date. Forward-filling a closed market also suppresses measured risk.

PYTHON / SYSTEM IMPLEMENTATION

Use exchange calendars, UTC timestamps, explicit session IDs, maximum-age rules, and tests for half-days and daylight changes. Persist whether a value is fresh, stale, or unavailable.

CONFUSIONS TO AVOID

Inner-joining calendars removes hard periods and silently changes the sample. Row shifting on an irregular panel does not prove causal alignment.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Calendar and freshness services support feature windows, scheduling, fills, valuation, risk, and live replay. Their versions belong in every backtest manifest.

RELATED CONCEPTS AND QUICK RECALL

Related: asynchronous markets, stale prices, session calendars, as-of alignment; these govern lawful cross-market comparison. Recall task: Design a causal rebalance using US equities, European futures, and an Asian macro release on a holiday week.

Target and preprocessing leakage

INTERMEDIATE

ONE-SENTENCE MEANING

Target leakage lets future outcomes influence features, transformations, selection, or fitted state. Preprocessing leakage occurs when scaling, imputation, winsorization, or encoding is fit beyond the training boundary.

MEMORY JOGGER

A feature can contain no explicit future column and still leak through how it was prepared. Fit every learned transform inside each temporal fold.

WHY IT MATTERS IN QUANT RESEARCH

Leakage produces stable-looking accuracy because future distribution information cleans the past. It invalidates both model comparison and uncertainty estimates.

FORMULA INTUITION

For fold k, transform state phi_k = fit(X_train_k) and X_test_k' = apply(phi_k, X_test_k). Label windows that intersect test periods must be purged from training.

SIMPLE MARKET EXAMPLE

Full-sample median imputation uses values from a later regime to fill earlier gaps. Selecting features by correlation on the entire sample leaks test labels before modeling begins.

PYTHON / SYSTEM IMPLEMENTATION

Build fit and transform interfaces with fold-scoped artifact IDs and forbid global learned state. Use synthetic future perturbations to confirm earlier features and choices remain unchanged.

CONFUSIONS TO AVOID

Chronological splitting after global preprocessing does not restore independence. Unsupervised transforms can still leak future distribution shifts even without labels.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Fold isolation spans data preparation, feature selection, model fitting, calibration, threshold choice, and evaluation. The artifact registry records state for every outer fold.

RELATED CONCEPTS AND QUICK RECALL

Related: target leakage, fold isolation, purging, fitted transforms; these prevent future outcomes from shaping historical inputs. Recall task: Audit a standardization-plus-feature-selection pipeline and mark every operation that must be refit within a fold.

Same-bar and close-price bias

INTERMEDIATE

ONE-SENTENCE MEANING

Same-bar bias grants a strategy a price from the interval used to form its decision. The error is common when bar summaries hide the ordering of observation, computation, routing, and execution.

MEMORY JOGGER

You cannot know a completed bar and trade inside that same completed bar without a specific auction or intrabar model. Move the fill to the first lawful window.

WHY IT MATTERS IN QUANT RESEARCH

Short-horizon edge can vanish between a signal timestamp and attainable execution. Correct sequencing separates prediction from accidental use of the answer price.

FORMULA INTUITION

Require observation_end <= data_ready < decision_time < order_arrival <= fill_time. The forward label begins at the simulated fill, not the bar used by the feature.

SIMPLE MARKET EXAMPLE

A close-to-close reversal signal computed from today's closing auction cannot buy at that same official close. A next-open fill must include overnight movement and auction costs.

PYTHON / SYSTEM IMPLEMENTATION

Represent bars with interval boundaries and model compute and route latency explicitly. Add invariants rejecting fills that precede their parent decision or order.

CONFUSIONS TO AVOID

A mechanical shift by one row can be too optimistic or too conservative when bars are irregular. The correct lag comes from event clocks and the execution venue.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Clock contracts connect source ingestion, feature readiness, scheduler, order simulator, label builder, and live latency monitoring. One event model should serve both replay and production.

RELATED CONCEPTS AND QUICK RECALL

Related: lookahead bias, decision cutoff, auction execution, label horizon; these define the first tradable price. Recall task: Place observation, decision, order, fill, and label times for a signal computed from the 16:00 closing print.

Fill, queue, and partial-execution bias

INTERMEDIATE

ONE-SENTENCE MEANING

Fill bias assumes orders execute more completely or favorably than historical market conditions support. Queue position, latency, volume ahead, cancellations, and adverse selection govern actual execution.

MEMORY JOGGER

A touched limit is not a guaranteed fill. Execution is a path with state, not a favorable lookup from bar high and low.

WHY IT MATTERS IN QUANT RESEARCH

Optimistic fills can manufacture alpha, especially for intraday, limit-order, and illiquid strategies. Partial execution also changes later holdings, risk, and orders.

FORMULA INTUITION

Market fill approximates arrival mid plus spread and impact; limit fill needs a trade-through or queue rule. Filled quantity cannot exceed available contra volume after participation and priority assumptions.

SIMPLE MARKET EXAMPLE

A buy limit at 100 touches a low of 99.99 but only 200 shares trade while 5,000 were ahead. Granting a full fill and capturing the rebound creates free optionality.

PYTHON / SYSTEM IMPLEMENTATION

Use quote, trade, or book data matched to horizon and persist open-order state. Stress queue priority, latency, participation, cancellation, and fill probability by market state.

CONFUSIONS TO AVOID

Random slippage without conditioning on order and market state adds noise rather than realism. Next-open execution can also be optimistic during gaps and auctions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The execution simulator consumes orders and market state, emits fills to the ledger, and reports shortfall and unfilled opportunity. Live fills should calibrate and challenge it.

RELATED CONCEPTS AND QUICK RECALL

Related: implementation shortfall, queue priority, adverse selection, participation; these translate intent into attainable holdings. Recall task: Write a conservative fill rule for a passive order and explain how an unfilled remainder affects the next rebalance.

Cost, impact, and capacity bias

INTERMEDIATE

ONE-SENTENCE MEANING

Cost bias understates spread, fees, impact, financing, borrow, delay, and missed trades. Capacity bias scales paper returns without recomputing the trades and market response required by larger capital.

MEMORY JOGGER

Gross edge is not scalable edge. Rebuild orders, fills, holdings, and risk at each proposed capital level.

WHY IT MATTERS IN QUANT RESEARCH

Small forecast advantages are often consumed by nonlinear implementation burden. Capacity analysis distinguishes a valid small strategy from an invalid large allocation.

FORMULA INTUITION

Net edge(Q) = gross edge - spread - fees - impact(Q) - financing - opportunity loss. Impact generally rises with participation and urgency rather than linearly with dollars.

SIMPLE MARKET EXAMPLE

An order uses 2 percent of daily volume at $1 million but 20 percent at $10 million. The larger book trades longer, moves prices, and holds stale targets during execution.

PYTHON / SYSTEM IMPLEMENTATION

Calibrate cost components by asset, side, volatility, size, session, and participation; retain conservative fallbacks. Sweep capital and stress common exits under thinner depth.

CONFUSIONS TO AVOID

Subtracting a fixed basis-point charge from strategy returns ignores turnover and trade direction. Using today's liquidity for old dates introduces a different form of lookahead.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Cost services feed optimization, trade suppression, execution simulation, capacity, net attribution, and live monitoring. Predicted-versus-realized shortfall updates require governance.

RELATED CONCEPTS AND QUICK RECALL

Related: bid-ask spread, market impact, turnover, liquidation horizon; these determine whether paper alpha survives implementation. Recall task: Decompose one round trip into every cost component, then explain which components change nonlinearly with capital.

SIMULATION AND ECONOMICS

Portfolio state and accounting drift

INTERMEDIATE

ONE-SENTENCE MEANING

Accounting drift occurs when targets, orders, fills, holdings, cash, and NAV are collapsed or updated inconsistently. It can create exposure and wealth without a corresponding economic event.

MEMORY JOGGER

Signal is belief, target is intent, order is instruction, fill is fact, and holding is exposure. Reconcile each transition before calculating performance.

WHY IT MATTERS IN QUANT RESEARCH

State errors often survive visual inspection because portfolio-level returns still look smooth. A balanced ledger exposes double dividends, duplicate fills, stale orders, and incorrect short liabilities.

FORMULA INTUITION

$NAV_t = cash_t + \sum_i q_{i,t} P_{i,t} + receivables_t - liabilities_t$. Quantity changes must equal fills plus corporate actions, and unexplained reconciliation residual must be zero.

SIMPLE MARKET EXAMPLE

A target rises from 100 to 300 shares, an order for 120 is sent, and 70 fill. The next decision begins from 170 shares plus any live remainder, not from the 300-share target.

PYTHON / SYSTEM IMPLEMENTATION

Use immutable economic events, idempotency keys, position snapshots, and daily ledger reconciliation. Fail the run when cash, quantity, or NAV residual exceeds a tight tolerance.

CONFUSIONS TO AVOID

Weighted-return arithmetic can hide state mistakes when leverage, intraday flows, or financing vary. Short-sale proceeds are cash paired with a liability, not free capital.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ledger is authoritative for exposure, returns, costs, attribution, and broker reconciliation. Research, paper, and live systems should consume the same event schema.

RELATED CONCEPTS AND QUICK RECALL

Related: event sourcing, self-financing identity, idempotency, reconciliation; these prevent wealth from appearing without cause. Recall task: Trace a partially filled rebalance through open orders, positions, cash, marks, and the next decision state.

Data snooping and multiple testing

INTERMEDIATE

ONE-SENTENCE MEANING

Data snooping repeatedly queries the same outcomes until a favorable rule appears. Multiple testing makes the best observed statistic optimistic because it was selected from a family of alternatives.

MEMORY JOGGER

The final backtest inherits every result that influenced its choice. Count abandoned variants, visual inspections, filters, and stopping decisions.

WHY IT MATTERS IN QUANT RESEARCH

Without search accounting, ordinary noise can look like a rare discovery and standard p-values become misleading. The remedy combines economic priors, logging, nested selection, and new evidence.

FORMULA INTUITION

For m independent null tests at level alpha, family false-positive probability is 1 - (1-alpha)^m. Dependence changes the exact number but does not erase selection pressure.

SIMPLE MARKET EXAMPLE

One hundred null signals tested at 5 percent produce about five nominal discoveries on average. Publishing only the best Sharpe hides the family that created it.

PYTHON / SYSTEM IMPLEMENTATION

Log all runs and define research families before ranking candidates. Use resampling, false-discovery controls, deflated performance, and sealed outer tests appropriate to dependence.

CONFUSIONS TO AVOID

Bonferroni alone does not repair leakage, bad fills, or a vague hypothesis. Correlated variants still contribute to search even when they are not one hundred independent bets.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment tracker, hypothesis registry, validation service, and promotion review share one immutable search ledger. Unregistered result shopping should block approval.

RELATED CONCEPTS AND QUICK RECALL

Related: family-wise error, false discovery rate, deflated Sharpe, nested validation; these price the search that produced a winner. Recall task: Estimate the family behind a chosen lookback and state which evidence remains independent after the choice.

Parameter mining and edge overfit

INTERMEDIATE

ONE-SENTENCE MEANING

Parameter mining chooses settings that fit historical noise rather than a stable mechanism. Edge overfit appears as a sharp optimum, fragile sign, or performance concentrated in a few observations.

MEMORY JOGGER

Prefer coherent plateaus to isolated peaks. A plausible mechanism should tolerate nearby settings and reasonable implementation perturbations.

WHY IT MATTERS IN QUANT RESEARCH

Highly optimized parameters raise turnover, model risk, and the chance of live disappointment. Stability surfaces show whether the rule expresses a durable effect or historical coincidence.

FORMULA INTUITION

Selection should compare out-of-sample utility over a neighborhood, not maximize one in-sample score. Penalize complexity, turnover, drawdown, and unstable derivatives of performance with respect to parameters.

SIMPLE MARKET EXAMPLE

A 63-day lookback works while 60 and 66 days fail despite identical economic logic. The spike is more consistent with sparse events or search noise than a precise market horizon.

PYTHON / SYSTEM IMPLEMENTATION

Publish complete parameter surfaces with support, cost, exposure, and fold dispersion. Select simple representatives inside training and evaluate the frozen choice once outside.

CONFUSIONS TO AVOID

A broad plateau is supportive but not proof because all cells may share leakage or beta. Robustness is meaningful only after causal and economic validity pass.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Parameter diagnostics feed model selection, uncertainty, risk limits, and change control. Live monitoring should test whether operation remains inside the validated neighborhood.

RELATED CONCEPTS AND QUICK RECALL

Related: sensitivity analysis, regularization, stability plateau, specification curve; these reveal how much the result depends on tuning. Recall task: Interpret a single-cell performance spike and design three perturbations that would distinguish mechanism from noise.

Selective reporting and narrative bias

INTERMEDIATE

ONE-SENTENCE MEANING

Selective reporting emphasizes favorable metrics, periods, subgroups, or stories while suppressing adverse evidence. Narrative bias then explains the chosen result as if the explanation preceded the search.

MEMORY JOGGER

Show the full scorecard and the full sample. Separate ex-ante mechanism from ex-post interpretation.

WHY IT MATTERS IN QUANT RESEARCH

Readers cannot calibrate a claim when losses, failed variants, and unstable regimes are absent. Complete reporting also preserves negative knowledge for future research.

FORMULA INTUITION

A report should bind primary metrics and slices before evaluation, then disclose all material deviations. Post-hoc analyses remain exploratory until independently replicated.

SIMPLE MARKET EXAMPLE

A strategy is described as defensive after performing well in one crisis, though the label was invented after viewing that episode. Other crises where it failed are omitted.

PYTHON / SYSTEM IMPLEMENTATION

Generate standard reports automatically with return, drawdown, tails, exposure, turnover, costs, capacity, support, uncertainty, and every declared stress slice. Archive rejected variants and reasons.

CONFUSIONS TO AVOID

More charts do not guarantee transparency if only favorable charts are chosen. A compelling economic story cannot convert reused evidence into a fresh test.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Reporting policy connects experiment tracking, validation, model cards, review, and knowledge retention. Promotion packets should contain mandatory adverse evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: publication bias, hindsight narrative, cherry-picking, negative results; these govern what evidence remains visible. Recall task: Rewrite one post-hoc market story as a preregistered hypothesis with explicit falsifiers and mandatory counterexamples.

Benchmark and exposure confounding

INTERMEDIATE

ONE-SENTENCE MEANING

Confounding occurs when a strategy's apparent alpha is explained by market, sector, style, volatility, liquidity, or timing exposure. A weak benchmark can make repackaged beta look like unique skill.

MEMORY JOGGER

Ask what simpler portfolio earns the same risk. Compare net results under matched universe, timing, leverage, turnover, and costs.

WHY IT MATTERS IN QUANT RESEARCH

Exposure-aware benchmarks reveal whether complexity adds forecast value or merely changes risk packaging. They also make capital allocation and drawdown expectations more honest.

FORMULA INTUITION

Candidate return = benchmark return + factor exposures times factor returns + residual. The residual is not automatically tradable alpha unless the hedge and costs are feasible.

SIMPLE MARKET EXAMPLE

A long-only value model beats cash but trails a cheap value index after matching beta and sector weights. Its complex score may add no incremental decision value.

PYTHON / SYSTEM IMPLEMENTATION

Build a benchmark ladder: cash or passive, exposure-matched, naive rule, prior production model, and randomized controls. Use identical accounting and execution assumptions.

CONFUSIONS TO AVOID

Regressing returns on factors after the fact can hide dynamic nonlinear exposures. Comparing a costed candidate with an uncosted baseline biases the decision in the opposite direction.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Benchmark services feed validation, attribution, risk budgeting, capital allocation, and retirement. Every promotion states the incremental capability purchased.

RELATED CONCEPTS AND QUICK RECALL

Related: factor attribution, matched baseline, incremental alpha, null model; these separate unique value from familiar risk. Recall task: Construct four benchmarks for a new cross-sectional model and specify how each controls a different confounder.

STATISTICAL SEARCH

Dependence and effective sample size

INTERMEDIATE

ONE-SENTENCE MEANING

Backtest observations share market shocks, holdings, labels, and regimes, so row count overstates independent evidence. Effective sample size is determined by distinct decisions and episodes, not database size.

MEMORY JOGGER

Count independent time support before quoting precision. Thousands of asset rows on one date may represent one common market event.

WHY IT MATTERS IN QUANT RESEARCH

Ignoring dependence narrows intervals, inflates t-statistics, and contaminates random train-test splits. Correct uncertainty changes which strategies deserve further research.

FORMULA INTUITION

A rough time-series approximation is n_eff = n / (1 + 2 sum_k rho_k). HAC errors, clustered inference, or block bootstrap should match the dependence structure.

SIMPLE MARKET EXAMPLE

A monthly signal held for twelve months creates overlapping portfolios with shared returns. Treating 240 monthly rows as independent substantially exaggerates evidence.

PYTHON / SYSTEM IMPLEMENTATION

Store label intervals, decision cohorts, asset groups, and regime blocks. Report date breadth separately from time depth and resample contiguous blocks or economic episodes.

CONFUSIONS TO AVOID

Clustering only by asset misses common-date shocks, while clustering only by date may miss serial overlap. More complex standard errors cannot repair lookahead or selection bias.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Dependence metadata feeds fold construction, statistical tests, model comparison, risk, and performance reporting. The strategy contract declares the unit of evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: autocorrelation, clustered errors, block bootstrap, overlapping labels; these determine the information content of a sample. Recall task: Estimate independent support for a twelve-month holding strategy tested monthly across five hundred stocks.

Temporal validation, purge, and embargo

INTERMEDIATE

ONE-SENTENCE MEANING

Temporal validation preserves the direction of information by fitting on earlier decisions and evaluating later ones. Purge and embargo remove training events whose labels or information overlap the test boundary.

MEMORY JOGGER

Split events by their full information and outcome intervals, not by row timestamp alone. Advance the frontier without shuffling time.

WHY IT MATTERS IN QUANT RESEARCH

Proper folds estimate how the research procedure handles changing markets while preventing shared future returns from crossing boundaries. They are essential for overlapping labels and long holdings.

FORMULA INTUITION

Training event i is excluded when its label interval intersects a test interval; embargo adds a post-test gap for lingering dependence. All preprocessing is fit only on retained training data.

SIMPLE MARKET EXAMPLE

A feature dated December has a twelve-month target extending through next November. It cannot remain in training when the test begins in January because most of its outcome is test-period information.

PYTHON / SYSTEM IMPLEMENTATION

Represent samples as intervals, generate walk-forward folds deterministically, and unit-test every boundary. Cache fold-scoped transforms and aggregate test returns in chronological order.

CONFUSIONS TO AVOID

Random cross-validation leaks regimes and overlapping outcomes even when rows are numerous. Purge does not justify using final-vintage data or globally fitted features.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The fold builder orchestrates datasets, transformations, model selection, backtests, and artifact registration. Production outcomes become forward evidence only after labels mature.

RELATED CONCEPTS AND QUICK RECALL

Related: walk-forward testing, label intervals, embargo, fold isolation; these protect chronological evaluation. Recall task: Draw a three-fold schedule for a six-month label and mark which training events must be removed around each test.

Holdout exhaustion and pseudo out-of-sample

INTERMEDIATE

ONE-SENTENCE MEANING

A holdout is exhausted when repeated inspection influences subsequent choices, turning it into development data. Calling the same period out-of-sample does not restore independence.

MEMORY JOGGER

Every peek spends evidence. Freeze the procedure before opening the final period and record what decisions remain allowed afterward.

WHY IT MATTERS IN QUANT RESEARCH

Repeated tuning to a public or internal benchmark creates selection optimism that ordinary test metrics omit. Honest governance distinguishes exploratory, validation, sealed, and live-forward evidence.

FORMULA INTUITION

Evidence remains confirmatory only while candidate choices are independent of its outcomes. After a result changes the procedure, future evaluation requires genuinely new time, market, or preregistered replication.

SIMPLE MARKET EXAMPLE

A team checks 2024 performance, adjusts a threshold, then calls 2024 out-of-sample in the report. The period helped choose the model and is therefore development evidence.

PYTHON / SYSTEM IMPLEMENTATION

Gate access to sealed metrics, log openings, require signed freeze records, and expose only pass-fail summaries when practical. Maintain a calendar for label maturation and forward reviews.

CONFUSIONS TO AVOID

Creating a new holdout from an already explored history only renames reused evidence. Cross-validation does not create independent data when the entire procedure was repeatedly adapted.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Evidence governance connects experiment roles, access controls, review boards, model registry, and forward monitoring. Promotion should disclose every period previously viewed.

RELATED CONCEPTS AND QUICK RECALL

Related: sealed test, adaptive overfitting, leaderboard leakage, live-forward evidence; these describe the consumption of a holdout. Recall task: Classify each historical period after three rounds of tuning and identify what genuinely independent evidence remains.

Robustness and failure surfaces

INTERMEDIATE

ONE-SENTENCE MEANING

Robustness testing maps performance and controls across plausible data, timing, parameter, cost, regime, and implementation changes. A failure surface shows where the mechanism stops being economically useful or operationally safe.

MEMORY JOGGER

Do not ask whether one backtest survives everything. Ask which assumptions carry the result and whether limits are observable before capital is harmed.

WHY IT MATTERS IN QUANT RESEARCH

Structured stress separates stable mechanism from one fragile path and defines an operating envelope for deployment. It also identifies adverse combinations history rarely contains.

FORMULA INTUITION

Evaluate utility(theta, state, capital) over a declared grid and retain support, uncertainty, exposure, turnover, and loss alongside return. Joint stresses matter because costs and risk interact.

SIMPLE MARKET EXAMPLE

A strategy tolerates double spread or one-day delay separately but fails when both occur during a volatility shock. The joint surface reveals a hidden margin problem.

PYTHON / SYSTEM IMPLEMENTATION

Automate perturbation suites, preserve complete surfaces, and predefine severity levels and failure thresholds. Include vendor alternatives, stale inputs, outages, and common liquidation.

CONFUSIONS TO AVOID

Randomly changing assumptions without economic calibration is not a meaningful stress test. Robustness cannot compensate for an invalid causal timeline.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Stress outputs feed promotion limits, capital sizing, monitoring thresholds, fallback, and retirement. Live state should be compared with the validated envelope.

RELATED CONCEPTS AND QUICK RECALL

Related: sensitivity analysis, scenario testing, operating envelope, margin of safety; these convert assumptions into explicit limits. Recall task: Design a joint stress grid for delay, spread, volatility, and liquidity, then define the first hard-stop boundary.

Reproducibility, parity, and change governance

INTERMEDIATE

ONE-SENTENCE MEANING

Reproducibility regenerates the same decisions and economics from immutable artifacts. Parity proves that research, paper, and live paths implement the same semantics under equivalent inputs.

MEMORY JOGGER

Same code is not enough when data, calendars, environments, defaults, or state differ. Compare boundary outputs and govern every semantic change.

WHY IT MATTERS IN QUANT RESEARCH

Without reproducibility, a defect cannot be isolated or repaired with confidence. Without parity, a valid backtest can still deploy as a different strategy.

FORMULA INTUITION

Run identity hashes code, parameters, data vintages, environment, calendars, seeds, and dependency graph. Parity tests compare features, targets, orders, holdings, and ledger events at exact boundaries.

SIMPLE MARKET EXAMPLE

A library upgrade changes ranking ties and live turnover rises while aggregate signals look similar. Boundary hashes locate the divergence before P&L becomes the first alarm.

PYTHON / SYSTEM IMPLEMENTATION

Use immutable artifacts, deterministic fixtures, container locks, golden event sequences, shadow replay, and semantic diffs. Route material changes through revalidation proportional to impact.

CONFUSIONS TO AVOID

Bitwise equality is not always required for numerical models, but tolerances must be declared and economically justified. Re-running a mutable query is not reproducibility.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Artifact registry, CI, replay, model registry, deployment, monitoring, incident response, and rollback form one control loop. Every live decision carries version and health state.

RELATED CONCEPTS AND QUICK RECALL

Related: artifact lineage, golden tests, shadow mode, semantic versioning; these keep the deployed procedure equal to the tested one. Recall task: Name the artifacts and boundary comparisons needed to prove a backtest and live engine make the same decision.

CONCEPT MAP

How research errors propagate into capital

INTERMEDIATE

READING RULE

Errors compound downstream. A small source defect can alter features, targets, fills, and reported risk, while a polished final metric hides the original break.

SOURCE EVIDENCE

vintage, identity, membership, actions, availability

DECISION DATASET

as-of joins, clocks, preprocessing, labels, folds

RESEARCH CHOICE

hypothesis, variants, stopping, benchmark, reporting

SIMULATED ECONOMICS

orders, queue, fills, costs, holdings, ledger

VALIDATION CLAIM

dependence, uncertainty, robustness, holdout role

CAPITAL SYSTEM

parity, limits, monitoring, incidents, retirement

AUDIT ACTION

Locate the earliest divergence, invalidate all dependent results, repair the boundary, add a permanent regression test, and regenerate descendants from immutable inputs.

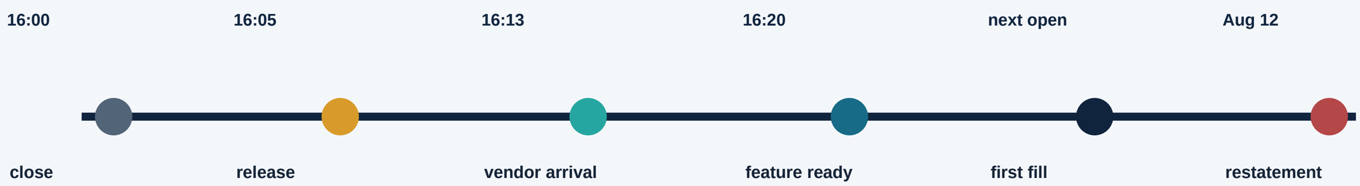
TIMELINE

Point-in-time clock and leakage boundary

INTERMEDIATE

SCENARIO

A company reports after the close, the vendor delivers later, and a restatement arrives months afterward. Each event creates a different lawful knowledge interval.



INVALID CLOSE TRADE

The release and processed feature did not exist by 16:00. A close fill uses information from after the traded price.

VALID NEXT-SESSION PATH

The feature is ready at 16:20, so the strategy may schedule the next lawful order. Overnight movement and auction costs belong in the test.

VINTAGE INTERVAL

The original value governs decisions until the August restatement becomes available. Historical decisions must not inherit the corrected number.

MUTATION TEST

Changing data after a cutoff must not alter earlier features or decisions. Later fills may change only through market state after their parent orders.

SYSTEM RECORD

Store event, publication, arrival, readiness, decision, order, fill, and revision times. Calendar and source versions complete the causal contract.

WORKED CALCULATION

Worked survivorship-bias calculation

INTERMEDIATE

SCENARIO

Five equally weighted stocks start the year. One delists at an 80 percent loss, but a current-survivor database silently removes it from the historical test.

Security	Total return	Year-end state
A	+18%	survives
B	+10%	survives
C	+6%	survives
D	-4%	survives
E	**-80%**	**delists**

SURVIVOR-ONLY RESULT

Removing E leaves average return = (18 + 10 + 6 - 4) / 4 = +7.5 percent. The sample now conditions on a future outcome and presents a profitable universe.

POINT-IN-TIME RESULT

Including every asset eligible at the start gives (18 + 10 + 6 - 4 - 80) / 5 = -10.0 percent. One omitted terminal event changes the sign and economic conclusion.

CROSS-SECTIONAL CONSEQUENCE

Deleting E also changes ranks, breadth, turnover, and weights before return averaging. Survivor bias therefore contaminates both the tested opportunity set and portfolio construction.

CONTROL

Snapshot historical eligibility before feature calculation and retain delisting proceeds, identity changes, and reasons for missingness. Compare vendor rosters against listing and terminal-event counts.

DIAGNOSTIC MATRIX

Leakage routes by research stage

INTERMEDIATE

USE

A lagged feature column can still be contaminated elsewhere. Audit every stage for future values, future fitted state, future selection, and overlapping outcomes.

	Future value	Future state	Future choice	Overlap
Source	AUDIT	AUDIT	LOWER	LOWER
Universe	AUDIT	AUDIT	AUDIT	LOWER
Join	AUDIT	AUDIT	LOWER	LOWER
Preprocess	LOWER	AUDIT	AUDIT	LOWER
Feature select	LOWER	AUDIT	AUDIT	AUDIT
Label	AUDIT	LOWER	LOWER	AUDIT
Fold	LOWER	AUDIT	AUDIT	AUDIT
Benchmark	AUDIT	LOWER	AUDIT	AUDIT

INTERPRETATION

Red cells identify common routes, not proof of contamination. Source vintages, current-survivor rosters, global transforms, feature screening, label overlap, and adaptive benchmarks require distinct tests.

CONTROL SET

Combine cutoff mutation tests, fold-scoped state, point-in-time rosters, interval-aware purge, sealed model selection, and benchmark preregistration. No single shift operation covers the matrix.

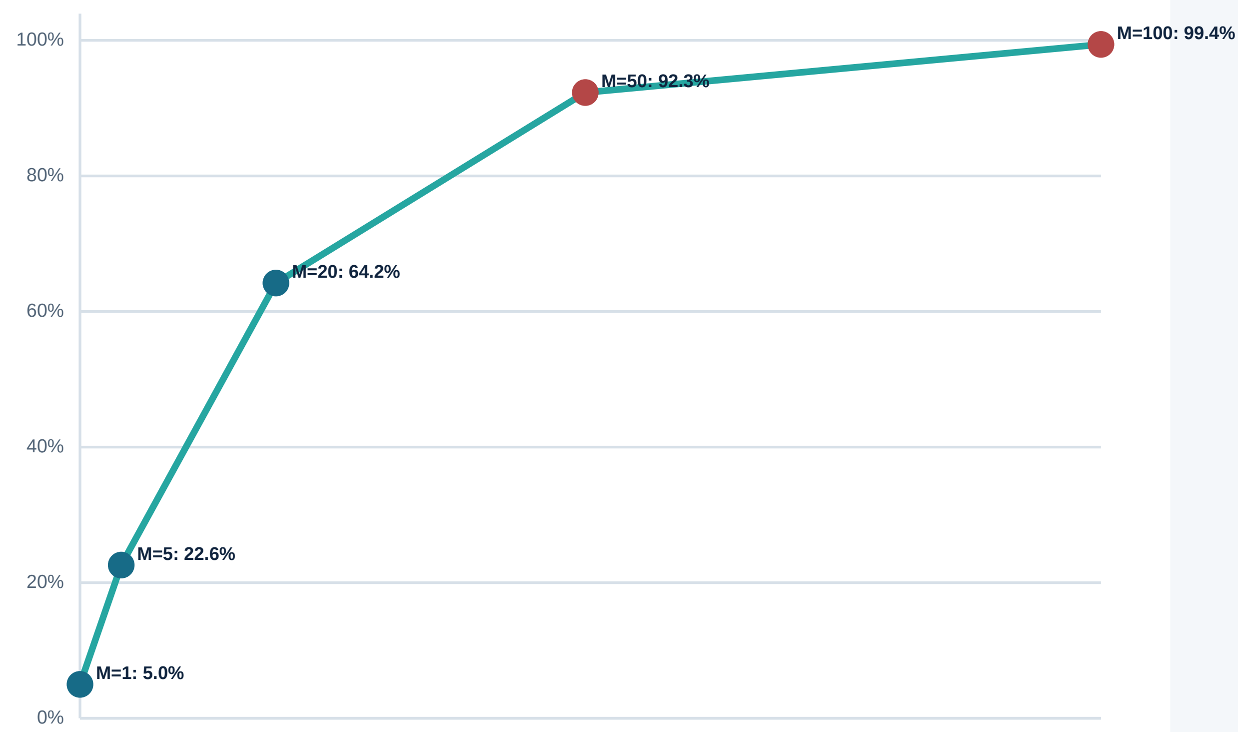
FORMULA AND WORKED EXAMPLE

Why repeated testing manufactures discoveries

INTERMEDIATE

EQUATION CALLOUT

Under independent null tests, P(at least one false positive) = 1 - (1 - alpha)^M. The expression measures family risk, not the probability that a specific surviving model is false.



WORKED INTERPRETATION

At alpha = 5 percent and M = 20 independent null trials, the chance of at least one nominal success is about 64.2 percent. At M = 100 it is about 99.4 percent.

REAL RESEARCH

Variants are dependent, so effective search size is smaller than a raw count but rarely equal to one. Preserve the family, use dependence-aware resampling, and demand sealed or forward replication.

BOUNDARY

Multiplicity adjustments address selection uncertainty only. They cannot legitimize future data, survivor bias, impossible fills, or an unreconciled ledger.

HEATMAP

Parameter surface: plateau or mined peak?

INTERMEDIATE

SYNTHETIC NET EVIDENCE

Cells combine outer-fold return, cost, drawdown, and support across lookback and threshold settings. The isolated best cell is less credible than a stable neighborhood.

	0.0	0.5	1.0	1.5	2.0
20d	FAIL	THIN	THIN	FAIL	FAIL
40d	THIN	THIN	GOOD	THIN	FAIL
60d	THIN	GOOD	SPIKE	THIN	FAIL
80d	THIN	GOOD	GOOD	THIN	FAIL
120d	FAIL	THIN	THIN	FAIL	FAIL

DIAGNOSIS

The 60-day, 1.0-threshold cell is an isolated spike rather than part of a broad stable region. Investigate sparse events, threshold discontinuities, exposure jumps, and trial selection.

SELECTION RULE

Choose a simple representative inside nested training using neighborhood utility and implementation margin. Evaluate that frozen choice once on sealed evidence and publish the complete surface.

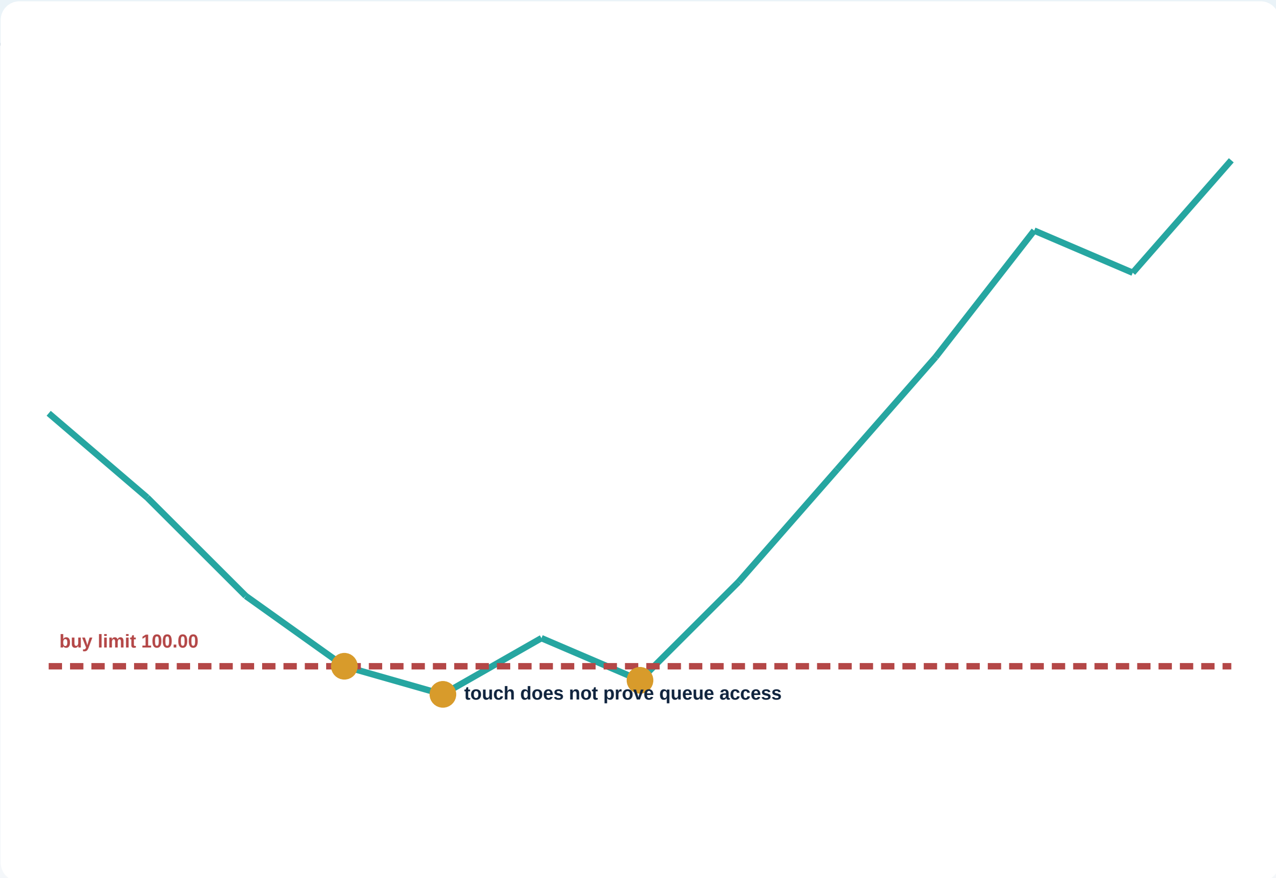
ANNOTATED CHART

Touched limits versus attainable fills

INTERMEDIATE

SYNTHETIC ORDER PATH

A passive buy limit is touched repeatedly, but queue depth and traded volume permit only partial execution. A bar-level full-fill rule overstates both exposure and rebound profit.



QUEUE RECONSTRUCTION

Suppose 5,000 shares are ahead, only 1,400 trade at or below the limit, and cancellations remove 3,900 ahead. At most 300 shares of a 1,000-share order can be justified before priority uncertainty.

BIAS MECHANISM

A full 1,000-share fill captures the entire rebound while ignoring seven hundred unfilled shares. The simulator grants favorable optionality precisely when liquidity was scarce.

CONSERVATIVE CONTROL

Require trade-through or queue evidence, enforce latency and participation, persist partial fills, and carry open-order state. Stress no-fill and adverse-selection cases beside the base model.

BIAS WATERFALL

Headline performance to credible evidence

INTERMEDIATE

SCENARIO

A strategy begins with a reported Sharpe of 2.1. Sequential repairs remove survivor advantage, same-bar execution, search optimism, and understated costs.

2.10	**Reported backtest**	selected headline before audit
-0.32	**Point-in-time universe**	restore failed and deleted securities
-0.28	**Lawful execution clock**	move fills behind decision and latency
-0.18	**Queue and partial fills**	remove touched-bar full-fill optionality
-0.24	**Full implementation cost**	spread, impact, fees, borrow, delay
-0.31	**Search adjustment**	price variants and holdout reuse
-0.17	**Dependence-aware estimate**	reduce pooled-row precision
0.60	**Credible working estimate**	still uncertain; requires forward evidence

INTERPRETATION

The waterfall is illustrative, not an additive estimator of universal bias. Its purpose is to show that independent-looking defects can compound and reverse the capital decision.

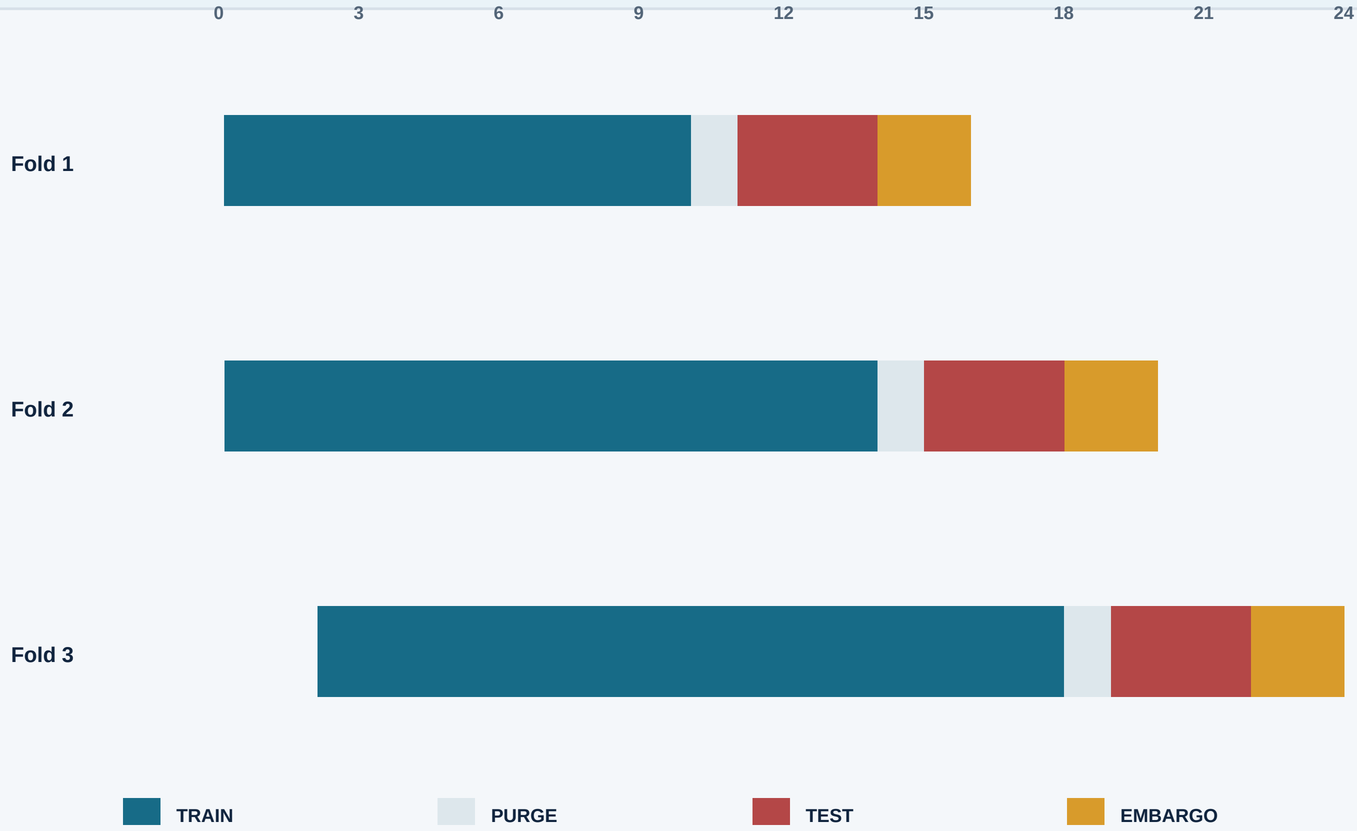
TIMELINE DIAGRAM

Walk-forward validation with purge and embargo

INTERMEDIATE

SETUP

Each event has a feature time and a future label interval. Training events that overlap the test outcome window are removed, and an embargo separates adjacent information states.



INTERVAL RULE

A training feature dated before the test may still be invalid if its label extends into the test window. Purging removes that overlap; embargo provides additional separation for lingering dependence or slow information propagation.

FOLD ISOLATION

Fit imputation, scaling, feature selection, model, calibration, and thresholds only on retained training events. Cache one artifact set per fold and never use test outcomes for repair.

EVIDENCE ROLE

Outer test results evaluate a frozen inner selection procedure. Once viewed and used to alter that procedure, the period becomes development evidence and a new forward test is required.

DIAGNOSTIC TABLE

Failure modes and first diagnostics

INTERMEDIATE

RESULTS IMPROVE AFTER A VENDOR REFRESH

Compare source vintages, rosters, adjustments, classifications, and timestamps. Earlier decisions may have inherited later corrections or survivor changes.

ONE PARAMETER DOMINATES EVERY CHART

Publish the full surface with support, turnover, exposure, cost, and fold dispersion. Inspect sparse events and threshold discontinuities before accepting the peak.

LIMIT STRATEGY LOOKS UNUSUALLY SMOOTH

Replay touched orders with latency, queue, trade-through, participation, partial fill, and adverse-selection assumptions. Compare no-fill opportunity cost.

SHARPE RISES WITH MORE ASSET ROWS

Measure independent dates, cohorts, label overlap, and common-factor shocks. Recalculate uncertainty with blocks or two-way clustering.

SHORT LEG CREATES NEARLY ALL PROFIT

Verify historical borrow, fees, recalls, dividends owed, terminal events, crowding, and forced cover. Remove unavailable shorts before neutrality is recomputed.

NAV CHANGES WITHOUT A LEDGER EVENT

Reconcile positions, cash, marks, actions, fees, liabilities, and external flows. Stop all performance reporting until the residual is zero.

OUT-OF-SAMPLE IMPROVES AFTER EACH REVISION

The test period is being consumed as development evidence. Freeze choices, disclose peeks, and wait for genuinely new labels or replication.

LIVE TURNOVER EXCEEDS THE BACKTEST

Compare data arrival, roster, fitted state, rounding, constraints, open orders, and scheduler outputs at each boundary. Treat the divergence as a parity incident.

REPAIR PRINCIPLE

Invalidate descendants from the earliest broken boundary, patch the cause, add a permanent regression test, and regenerate the evidence graph from immutable inputs.

PROMOTION GATE

Bias audit and promotion scorecard

INTERMEDIATE

QUESTION AND SEARCH

Mechanism, decision use, falsifiers, primary metrics, candidate family, manual choices, stopping rule, and all viewed results are registered.

POINT-IN-TIME EVIDENCE

Identity, universe, vintages, availability, actions, calendars, quality, liquidity, borrow, and missing states reproduce every historical decision.

FOLD ISOLATION

Preprocessing, selection, fitting, calibration, and thresholds are training-only; label intervals, purge, embargo, and sealed roles are verified.

EXECUTABLE SIMULATION

Decision, target, order, queue, fill, holding, cash, action, liability, and mark transitions are timed, conservative, and auditable.

ECONOMIC RECONCILIATION

Every quantity and cash change has an event; NAV, return, financing, costs, benchmark, and attribution identities balance exactly.

STATISTICAL EVIDENCE

Dependence, effective support, uncertainty, multiple testing, benchmarks, complete specification surfaces, and replication are disclosed.

IMPLEMENTATION MARGIN

Delay, spread, fees, impact, partial fills, borrow, financing, capacity, and stressed common exit leave sufficient net value.

ROBUST OPERATING ENVELOPE

Data, timing, parameter, regime, capital, and joint stresses define observable warning, de-risking, and stop boundaries.

REPRODUCIBILITY AND PARITY

Immutable artifacts, deterministic fixtures, boundary hashes, shadow replay, versioned environments, semantic diffs, and rollback are tested.

HARD STOP

Future data, survivor-only rosters, impossible fills, hidden search, exhausted holdouts, unreconciled NAV, or no rollback blocks capital.

INCIDENT RUNBOOK

Turning research defects into permanent controls

INTERMEDIATE

1. CONTAIN

Pause promotion or live capital for affected versions and preserve current artifacts. Do not overwrite the evidence needed to reconstruct the defect.

2. LOCATE

Trace the earliest divergent source, clock, transform, choice, order, fill, ledger, metric, or deployment boundary. Quantify every dependent run and decision.

3. REPRODUCE

Create the smallest deterministic fixture that triggers the failure. Bind source records, expected state transitions, and an explicit failing invariant.

4. REPAIR

Fix the causal boundary rather than tuning downstream outputs. Regenerate descendants and compare both intended and unintended semantic changes.

5. TEST

Convert the fixture into a permanent unit, integration, replay, or property test. Add control coverage to the relevant error class and owner.

6. REVALIDATE

Repeat selection and validation when the repair changes historical decisions or economics. Previously favorable metrics do not survive an invalidated evidence chain by default.

7. LEARN

Record cause, detection gap, affected scope, capital consequence, and prevention action. Preserve the failed artifact so future teams do not rediscover the same mistake.

FINAL REVIEW

Final active recall and system index

INTERMEDIATE

DEFINE

State the economic mechanism, decision use, falsifiers, complete procedure, primary metrics, research family, and stopping rule before judging results.

RECONSTRUCT

Freeze point-in-time identity, universe, vintages, actions, clocks, calendars, quality, liquidity, borrow, and every fitted artifact.

SIMULATE

Separate signal, target, order, queue, fill, holding, cash, liability, mark, and NAV. Reject transitions that precede their causes or fail conservation.

RECONCILE

Prove every quantity and cash change, cost, action, financing item, and return. Stop when the ledger or attribution has an unexplained residual.

PRICE SELECTION

Log variants, manual choices, holdout peeks, benchmarks, slices, and stopping decisions. Adjust uncertainty for dependence and the search that selected the winner.

VALIDATE

Use chronological folds, fold-scoped transforms, purge, embargo, matched baselines, complete surfaces, sealed evidence, and forward replication.

STRESS

Map data, timing, parameter, cost, capital, regime, outage, and common-exit assumptions jointly. Define observable operating limits and safety margin.

OPERATE

Bind immutable lineage, compare boundary parity, shadow decisions, monitor evidence and execution, rehearse rollback, and retire unsupported systems.

REJECT

Do not promote future knowledge, current survivors, same-bar prices, fantasy fills, fixed-cost scaling, isolated peaks, pseudo holdouts, or mutable reruns.

RETAIN

Store failed ideas, invalidated claims, incidents, negative replications, and regression fixtures. Durable research systems remember why an attractive result was rejected.